

Assignment - 4

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Aim : (Loops)

- Accept two numbers from the user (Minimum and Maximum). Find numbers which are divisible by 5 and multiple of 9 between the given range of minimum and maximum numbers.
- Write a Python program to count the number of prime, even and odd numbers from a series of numbers.

Theory : The aim for this assignment is to learn about loops and how they can be useful

There are 2 main types of Loops **for** and **while** in Python

Why would we need loops for this?

Loops are required because contents of array can not be inputted or displayed using sequential processes and a simple for loop is required to automate it to a certain extent

An example for **for** loop is :

```
fruits = ["apple", "banana", "cherry"]
for x in fruits:
    print(x)
```

Here, the variable x would go through all the element of the array fruits and print all the elements

An example for **while** loop is :

```
i = 1
while i < 6:
    print(i)
    i += 1
```

In this type of loop the process will continue till all criteria is satisfied in this case it would run till all numbers from 1 to 5 are printed until then the variable **i** would keep incrementing

Using these 2 basic loops and in built functions like **append** (as the name suggests append modifies the elements inside the array) we can modify elements of array or perform operations on them

Logic for finding numbers divisible by 5 and 9 both, would be that the number has to be divisible

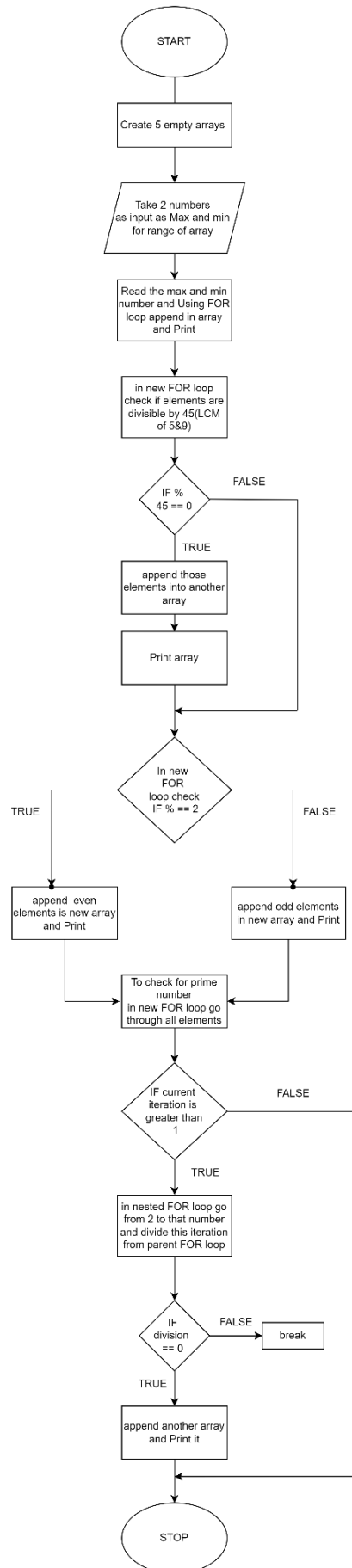
by both 5 & 9 so for the logical AND operations we use **and** in python where if and only if both the statements are true can the result be **True**

Logic for odd and even numbers, is that if the number when divided by 2 gives the result as 0 or not

If 0 then number is even otherwise odd

Logic for prime numbers, Prime number is the number which is divisible by 1 and itself only this is a bit of a complex logic to implement where nested loops would be required where the loop will pass through all the elements of the array and divide the current iteration into another for loop where it is divided by all numbers from 2 to the current number(minus 1) and if the result is 0 means the number in current iteration is divisible by some other number than 1 and itself and thus is a composite(non prime) number.

From our previous knowledge of asking for input from the user, declaring variables, creating if-else statements, and performing arithmetic operations we can easily create this program



Program :

```
## If-Else MENU based program by Rudradev Arya
a=[]
b=[]
c=[]
d=[]
prime=[]

n1 = int(input("Min number :: "))
n2 = int(input("Max number :: "))

inputnum = int(input("""\n
Enter 2 for displaying the Array,
Enter 3 for finding if array is divisible by both 5 & 9,
Enter 4 for Getting Even and Odd numbers from the array,
Enter 5 for getting Prime numbers from the array
:: """))

if inputnum == 2:
    for i in range(n1, n2):
        a.append(i)
    print("The array between min to max value is :: ", a)

elif inputnum == 3:
    for j in range(n1, n2):
        if j % 45 == 0:
            b.append(j)
    print("Elements from array divisible by 5 & 9 are :: ", b)

elif inputnum == 4:
    for k in range(n1, n2):
        if k % 2 == 0:
            c.append(k)

    else:
```

```

        d.append(k)

    print("even number are ", c)
    print("odd number are ", d)

elif inputnum == 5:
    for num in range(n1, n2):
        if (num > 1):
            for chk in range(2, num):
                if (num % chk) == 0:
                    break
            else:
                prime.append(num)

    print("List of prime numbers are :: ", prime)

else:
    print("Invalid Choice")

```

```

## Function based MENU based program by Rudradev Arya
a = []
b = []
c = []
d = []
prime = []

def menu():

    print()

print("=====
=====")

    print("Enter 1 for Inputting max & min")
    print("Enter 2 for displaying the Array")
    print("Enter 3 for finding if array is divisible by both 5 & 9")

```

```

print("Enter 4 for Getting Even and Odd numbers from the array")
print("Enter 5 for getting Prime numbers from the array")
print("Enter 0 to Exit")

print("=====")
print()

menu()
choice = int(input("Enter Your Choice :: "))

while(choice != 0):
    if choice == 1:
        n1 = int(input("Min number :: "))
        n2 = int(input("Max number :: "))

        elif choice == 2:
            for i in range(n1, n2):
                a.append(i)
            print("The array between min to max value is :: ", a)

        elif choice == 3:
            for j in range(n1, n2):
                if j % 45 == 0:
                    b.append(j)
            print("Elements from array divisible by 5 & 9 are :: ", b)

        elif choice == 4:
            for k in range(n1, n2):
                if k % 2 == 0:
                    c.append(k)

                else:
                    d.append(k)

            print("even number are :: ", c)
            print("odd number are :: ", d)

        elif choice == 5:

```

```

        for num in range(n1, n2):
            if (num > 1):
                for chk in range(2, num):
                    if (num % chk) == 0:
                        break
                else:
                    prime.append(num)

        print("List of prime numbers are :: ", prime)

    menu()
    choice = int(input("Enter Your Choice :: "))
else:
    print("Exiting!!!!!!!!")

```

Output :

```

## Output for If-Else MENU based program
Min number :: 11
Max number :: 22

Enter 2 for displaying the Array,
Enter 3 for finding if array is divisible by both 5 & 9,
Enter 4 for Getting Even and Odd numbers from the array,
Enter 5 for getting Prime numbers from the array
:: 2
The array between min to max value is ::  [11, 12, 13, 14, 15, 16, 17, 18,
19, 20, 21]

=====
=====

Min number :: 11
Max number :: 22

```

```
Enter 2 for displaying the Array,  
Enter 3 for finding if array is divisible by both 5 & 9,  
Enter 4 for Getting Even and Odd numbers from the array,  
Enter 5 for getting Prime numbers from the array  
:: 3  
Elements from array divisible by 5 & 9 are ::  []
```

```
=====
```

```
Min number :: 11  
Max number :: 22
```

```
Enter 2 for displaying the Array,  
Enter 3 for finding if array is divisible by both 5 & 9,  
Enter 4 for Getting Even and Odd numbers from the array,  
Enter 5 for getting Prime numbers from the array  
:: 4  
even number are  [12, 14, 16, 18, 20]  
odd number are  [11, 13, 15, 17, 19, 21]
```

```
=====
```

```
Min number :: 11  
Max number :: 22
```

```
Enter 2 for displaying the Array,  
Enter 3 for finding if array is divisible by both 5 & 9,  
Enter 4 for Getting Even and Odd numbers from the array,  
Enter 5 for getting Prime numbers from the array  
:: 5  
List of prime numbers are ::  [11, 13, 17, 19]
```

```
## Output for Function based MENU based program
```

```
=====
=====
Enter 1 for Inputting max & min
Enter 2 for displaying the Array
Enter 3 for finding if array is divisible by both 5 & 9
Enter 4 for Getting Even and Odd numbers from the array
Enter 5 for getting Prime numbers from the array
Enter 0 to Exit
=====
=====
```

```
Enter Your Choice :: 1
Min number :: 22
Max number :: 55
=====
=====
```

```
Enter 1 for Inputting max & min
Enter 2 for displaying the Array
Enter 3 for finding if array is divisible by both 5 & 9
Enter 4 for Getting Even and Odd numbers from the array
Enter 5 for getting Prime numbers from the array
Enter 0 to Exit
=====
=====
```

```
Enter Your Choice :: 2
The array between min to max value is :: [22, 23, 24, 25, 26, 27, 28, 29,
30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47,
48, 49, 50, 51, 52, 53, 54]
=====
=====
```

```
Enter 1 for Inputting max & min
Enter 2 for displaying the Array
Enter 3 for finding if array is divisible by both 5 & 9
Enter 4 for Getting Even and Odd numbers from the array
Enter 5 for getting Prime numbers from the array
Enter 0 to Exit
```

```
=====
=====
Enter Your Choice :: 3
```

```
Elements from array divisible by 5 & 9 are :: [45]
```

```
=====
=====
Enter 1 for Inputting max & min
```

```
Enter 2 for displaying the Array
```

```
Enter 3 for finding if array is divisible by both 5 & 9
```

```
Enter 4 for Getting Even and Odd numbers from the array
```

```
Enter 5 for getting Prime numbers from the array
```

```
Enter 0 to Exit
=====
=====
```

```
Enter Your Choice :: 4
```

```
even number are :: [22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46,
48, 50, 52, 54]
```

```
odd number are :: [23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43, 45, 47,
49, 51, 53]
```

```
=====
=====
Enter 1 for Inputting max & min
```

```
Enter 2 for displaying the Array
```

```
Enter 3 for finding if array is divisible by both 5 & 9
```

```
Enter 4 for Getting Even and Odd numbers from the array
```

```
Enter 5 for getting Prime numbers from the array
```

```
Enter 0 to Exit
=====
=====
```

```
Enter Your Choice :: 5
```

```
List of prime numbers are :: [23, 29, 31, 37, 41, 43, 47, 53]
```

```
=====
=====
Enter 1 for Inputting max & min
```

```
Enter 2 for displaying the Array
Enter 3 for finding if array is divisible by both 5 & 9
Enter 4 for Getting Even and Odd numbers from the array
Enter 5 for getting Prime numbers from the array
Enter 0 to Exit

=====

=====

Enter Your Choice :: 0
Exiting!!!!!!!
```

Conclusion : In this assignment we learnt how to implement loops, use arrays to create and store a range of numbers, find multiple of 5 & 9, odd, even and prime numbers using loops